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Conformal Prediction under High-Dimensional Covariate Shift

Guarantees in Machine Learning, Project Proposal. Supervisor: Francesco Taroni

Main paper and field

This project studies conformal prediction under covariate shift: how to keep prediction sets reliable when the model is deployed on inputs that differ from its calibration data. The main paper is *Conformal Inference under High-Dimensional Covariate Shifts via Likelihood-Ratio Regularization* by Joshi et al. 2025 [1].

Motivation

Conformal prediction turns any trained model's output into prediction sets that contain the true value with probability at least $1-\alpha$. Split conformal prediction, the most common variant, computes the model's errors on a held-out calibration set and takes their $(1-\alpha)$-quantile as the set width. The guarantee holds in finite samples and without assumptions on the model or the data distribution, requiring only that calibration and test points are exchangeable.

This assumption fails under covariate shift: the input distribution changes from $\mathbb{P}_1$ (calibration) to $\mathbb{P}_2$ (test) while the conditional $P(Y \mid X)$ remains the same. The model can be far less accurate on the test inputs, and coverage drops below the nominal level without warning. We are interested in the setting where a neural network surrogate replaces an expensive simulation inside a design optimizer. The conditional relationship does not change, but the optimizer explores inputs that differ from the training data, often in regions where the surrogate is least reliable, producing exactly the covariate shift setting with labeled calibration data from $\mathbb{P}_1$ and only unlabeled inputs from $\mathbb{P}_2$.

Three families of methods address this setting. The first reweights calibration errors by the likelihood ratio $r = d\mathbb{P}_2/d\mathbb{P}_1$, restoring exact coverage when r is known [2]. When r must be estimated, coverage degrades with the estimation error [3, 4]. An alternative fits a quantile regression over a function class, giving coverage under every shift in that class simultaneously without requiring target data [5], at the cost of over-coverage of order $\mathcal{O}(d/(n+1))$. A third family, distributionally robust methods, guarantees coverage under the worst-case shift within a neighborhood of the calibration distribution [6]. When inputs are high-dimensional (images, embeddings, physical geometries) and labeled data is expensive, all three face difficulties: density ratio estimation becomes unreliable, class-wide coverage is conservative, and robust neighborhoods require labeled test data to calibrate.

Paper's methodology

Joshi et al. address the case where inputs are high-dimensional and estimating r directly is impractical. Their method (LR-QR) fits an input-dependent threshold $h(x)$ through a set of features $\Phi(x)$ (for example the embedding of a pretrained network) by minimizing

$$\min_{h \in \mathcal{H},\, \beta \in \mathbb{R}} \; \mathbb{E}_1[\ell_\alpha(h(X), S)] + \lambda\big(\beta^2\, \mathbb{E}_1[h(X)^2] - 2\beta\, \mathbb{E}_2[h(X)]\big),$$

where S is the conformity score of a labeled calibration point and ℓ_α is the pinball loss, whose minimizer is the $(1-\alpha)$-quantile. The first term is a quantile regression on calibration data, as in [5], but instead of guaranteeing coverage under all shifts in the function class, the second term uses unlabeled inputs from $\mathbb{P}_2$ to focus on the specific shift that occurs. Since $\mathbb{E}_2[h] = \mathbb{E}_1[r\,h]$, this term equals the squared distance between βh and r up to a constant, steering the threshold towards the actual shift and avoiding the over-coverage penalty of the class-wide guarantee. At the population level, coverage at $1-\alpha$ holds assuming r lies in the span of the features, with additional estimation error and misspecification terms in finite samples. The paper validates the method on tabular data, an LLM question-answering task, and cell images with synthetic shifts, but does not consider physical simulation data.

The goal of this project is to clarify the relationship between LR-QR and the results it builds on, and to assess whether the method's assumptions and practical choices (feature selection, regularization strength, shift magnitude) transfer to application settings like surrogate-based design optimization, where the shift is induced by an optimizer and inputs are high-dimensional.

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Related Work

Conformal prediction under covariate shift [2]. Introduces the reweighting approach and proves exact coverage when the likelihood ratio is known. This is the starting point for the problem LR-QR addresses and one of its baselines.

Conformal prediction beyond exchangeability [7]. Extends conformal prediction to non-exchangeable data, bounding the coverage loss by how far the data depart from exchangeability. It provides the general framework for analysing methods under shift.

Conformal prediction with conditional guarantees [5]. Shows that quantile regression over a d-dimensional function class guarantees coverage under every shift in that class simultaneously, without target data, at the price of sets that over-cover by up to order $d/(n+1)$. LR-QR builds on this construction and adds unlabeled target inputs to focus on the shift that actually occurs.

A least-squares approach to direct importance estimation [8]. Estimates the likelihood ratio by minimizing $\frac{1}{2}\mathbb{E}_1[g^2] - \mathbb{E}_2[g]$, which up to a factor of two is the LR-QR regularizer with $g = \beta h$. The authors of LR-QR show that their fitted βh stays far from r; we want to understand what the regularizer actually does and how much of the method follows from these existing pieces.

Next Steps

1. Position LR-QR in the literature: what problem it addresses that earlier methods left open, and how later work relates to it.
2. Work through the paper's proofs, assumptions, and experiments, with LLM- and Lean-assisted checking.
3. Identify which assumptions and practical choices might not transfer to surrogate-based design: whether the features capture the shift, how the regularization strength is chosen, and how optimization-induced shifts are handled.
4. Select a concrete question from the above and develop it, combining theoretical analysis with small-scale experiments.

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